

Question	Marking Guidance	Mark	Comments
06.1	1. Used to compare effect of other treatments / as a baseline; 2. Shows / Measures effect of substance (X); OR Accounts for effect of substances produced naturally;	2	Accept for 2 marks, substance (X) and not agar / block / water that caused the difference in the number of roots. 1. Do not accept unqualified reference to "compare results". 2. Accept measures effect of independent variable
06.2	1. (D shows) substance (X) is not required for (some) root growth / production of roots; OR Substances (already) present in stem cause (some) root growth; 2. Substance X moves through plant; 3. (E shows) substance (X) causes / increases / doubles number of roots / root growth;	3	2. Accept X moves through stem / phloem
06.3	In support of mass flow hypothesis 1. (F shows) phloem is involved; 2. (G shows) respiration / active transport is involved (in flow / movement); 3. Because 4 °C / cooling reduces / slows / stops flow / movement; 4. The agar block is the source; 5. Roots are the sink; Against the mass flow hypothesis 6. No bulge above ringing (in F); 7. No (role for) osmosis / hydrostatic pressure / water movement; 8. Movement could be due to gravity; 9. Roots still grow without (intact/functioning) phloem; 10. No leaves / sugars / photosynthesis to act as a source;	4 max	Each point must be clearly made in the context of support or against. Ignore sugar / sucrose 3 max for "support" and 3 max for "against" 7. Accept no turgor pressure